Technical review

Item #8 · tsa.crt · Run ID: 20261113T232802Z-2f0fe5

Path	R:\evidence\certs\tsa.crt
Size (bytes)	1261
Last modified (UTC)	2026-11-12T06:00:00Z
Extension	.crt
Detected type (signature)	X.509 certificate
Status	No issues found
Case	RV-SAMPLE-001
Evidence ID	E0008
Case registry	Verified: identical to the registered evidence
Registered at intake (UTC)	2026-11-13T23:27:52.795Z

Hashes

sha256	`3159fd24ec2405866a88cda9728442102876a482038cfe0fdf24b1482799712c`
md5	`9fa8837d2e1780e53530f72a6c6b68db`

Findings

[Informational] Certificate chain verified at the date checked (CERT_CHAIN_VALID)

What it shows	OpenSSL built a path from C=CO, O=Rakvor Lab, CN=Rakvor Lab TSA to the trust anchor file:R:\anchors\rakvor-lab-root.crt and the chain verified at 2026-06-01T00:00:00Z: signatures, validity dates, purpose and constraints passed. Revocation is reported separately: a REVOCATION_UNKNOWN finding for the same certificate means its revocation status could not be determined.
What it does not prove	That the certificate was not revoked (see the revocation findings), that the certificate holder signed or approved any particular document, that the anchor deserves trust (that is the examiner's choice of --anchor), or that the result holds at another date.

Method

Each file was opened read-only, hashed in a single pass with the selected algorithms, and checked against a table of file signatures. Every action was written to a hash-chained audit log.

Audit trail

Audit entries	#34 – #37
Log folder	`R:\case\runs\rv-certcheck\20261113T232802Z-2f0fe5\logs`
Digital signature	This PDF is not digitally signed.

Limitations

- Every chain is judged at one instant (--at, recorded in each report); the same certificate can be valid at one date and expired or revoked at another.
- Offline, revocation is checked only against the CRLs and OCSP responses supplied (--crl, --ocsp, or found among the evidence); a revocation published elsewhere is not seen.

- Trust lists are used as dated snapshots (date and SHA-256 recorded); services added, suspended or withdrawn after the snapshot are not reflected.
- In version 0.1 the trust list's own XAdES signature is not verified; the snapshot's source and SHA-256 are recorded so it can be checked independently. The Colombian list is a curated file, because ONAC publishes no machine-readable list.
- Certificates issued by Rakvor's internal PKI are for internal use only: they are not a firma digital under Ley 527 de 1999, art. 28 (Colombia), and not qualified certificates under eIDAS (Regulation (EU) No 910/2014).
- The tool's results have been compared with OpenSSL and Python reference implementations on test sets, but no error rate has yet been measured on a validated corpus.
- A result without findings does not prove that a file is authentic or unaltered before acquisition.
- Modification times come from the file system of the examined copy and may differ from the original source.
- A trusted timestamp of the audit log shows that the log existed in this exact form at the stated time; it does not show that the examined copy matches the original source.